

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations November/December 2024

PRINCIPLES OF COMMUNICATION SYSTEMS

(Electrical & Electronics Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Illustrate the difference between DSB-SC and DSB-FC. 2M
 - (b) Write the main functions of a radio transmitter. 2M
 - (c) Draw the waveform of FM. 2M
 - (d) What is the advantage of FM demodulation over AM demodulation in terms of noise performance? 2M
 - (e) Define a band-pass signal and give an example. 2M
 - (f) How does oversampling affect the quality of a reconstructed signal? 2M
 - (g) What is the principle behind Binary Phase Shift Keying (BPSK)? 2M
 - (h) Discuss how does a regenerative repeater differs from an analog repeater. 2M
 - (i) List the role of the base station in mobile communication. 2M
 - (j) Mention the concept of IP addressing in computer networks. 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Demonstrate Double Sideband Full Carrier (DSB-FC) modulation, and derive its mathematical expression. 10M
- OR**
- 3 Describe the concept of frequency translation and why it is necessary in communication systems. Discuss the process of up-conversion and down-conversion with examples. 10M
- 4 Explain the concept of Angle Modulation. Compare Frequency Modulation (FM) and Phase Modulation (PM) in terms of their principles, waveforms, and applications. 10M
- OR**
- 5 Describe two different methods used for FM modulation. Also, explain the basic principle of FM demodulation and compare two demodulation techniques. 10M
- 6 Exemplify the concept of Pulse Amplitude Modulation (PAM) in detail. 10M
- OR**
- 7 Paraphrase the Sampling Theorem and its significance in signal processing. 10M
- 8 Describe the working of Binary Phase Shift Keying (BPSK) modulation. 10M
- OR**
- 9 What is a Regenerative Repeater and why is it important in digital communication systems? Elaborate. 10M
- 10 With the help of a block diagram, describe the working of an optical fiber communication system. Discuss the advantages of optical communication over traditional copper-based communication systems. 10M
- OR**
- 11 Construct the block diagram of a cellular mobile communication system. Discuss the working of different blocks and how the handoff process occurs during movement between cells. 10M
